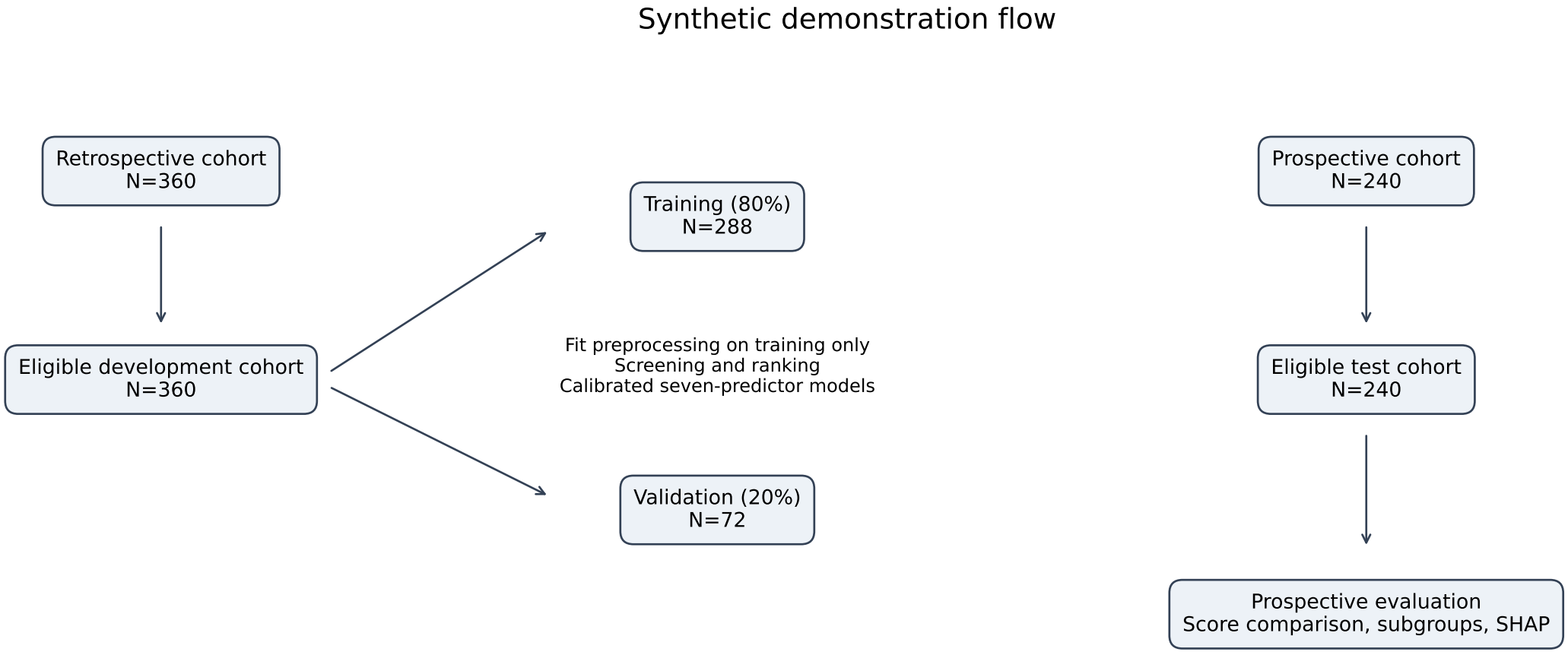
Synthetic demonstration flow
Retrospective cohort
N=360
Eligible development cohort
N=360
Training (80%)
N=288
Validation (20%)
N=72
Fit preprocessing on training only
Screening and ranking
Calibrated seven-predictor models
Prospective cohort
N=240
Eligible test cohort
N=240
Prospective evaluation
Score comparison, subgroups, SHAP